

NITRIC ACID (> 70% in water)**ICSC: 0183 (November 2016)****CAS #: 7697-37-2****UN #: 2031****EC Number: 231-714-2**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible but enhances combustion of other substances. Gives off irritating or toxic fumes (or gases) in a fire. Risk of fire and explosion on contact with incompatible substances. See Chemical Dangers.	NO contact with incompatible materials: See Chemical Dangers	Use water in large amounts, carbon dioxide. NO powder, foam. In case of fire: keep drums, etc., cool by spraying with water. NO direct contact of the substance with water.

AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Sore throat. Burning sensation. Shortness of breath. Laboured breathing.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest. Half-upright position. Artificial respiration may be needed. Refer immediately for medical attention.
Skin	Pain. Yellow staining of the skin. Serious skin burns.	Protective gloves. Protective clothing. Apron.	Wear protective gloves when administering first aid. First rinse with plenty of water for at least 15 minutes, then remove contaminated clothes and rinse again. Refer immediately for medical attention.
Eyes	Redness. Pain. Severe burns.	Wear face shield or eye protection in combination with breathing protection.	Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.
Ingestion	Burns in mouth and throat. Burning sensation behind the breastbone. Abdominal pain. Vomiting. Shock or collapse.	Do not eat, drink, or smoke during work.	Rinse mouth. Give nothing to drink. Do NOT induce vomiting. Refer immediately for medical attention.

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Evacuate danger area! Consult an expert! Personal protection: chemical protection suit including self-contained breathing apparatus. Do NOT absorb in saw-dust or other combustible absorbents. Ventilation. Collect leaking liquid in sealable containers. Cautiously neutralize remainder with sodium carbonate. Then wash away with plenty of water.	According to UN GHS Criteria  DANGER May be corrosive to metals May intensify fire; oxidizer May cause respiratory irritation Causes severe skin burns and eye damage See Notes
STORAGE	
Separated from combustible substances, reducing agents, bases, organic chemicals and food and feedstuffs. Cool. Dry. Keep in a well-ventilated room. Store only in original container.	
PACKAGING	
Unbreakable packaging. Put breakable packaging into closed unbreakable container. Do not transport with food and feedstuffs.	Transportation UN Classification UN Hazard Class: 8; UN Subsidiary Risks: 5.1; UN Pack Group: I



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NITRIC ACID (> 70% in water)**ICSC: 0183****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS-TO-YELLOW LIQUID WITH PUNGENT ODOUR.

Physical dangers

No data.

Chemical dangers

Decomposes on warming. This produces toxic and irritating fumes and gases including nitrogen oxides. The substance is a strong oxidant. It reacts violently with combustible and reducing materials, such as turpentine, charcoal and alcohol. The substance is a strong acid. It reacts violently with bases and is corrosive to metals. This produces flammable/explosive gas (hydrogen - see ICSC 0001). Reacts violently with organic compounds.

Formula: HNO₃

Molecular mass: 63.0

Boiling point: 121°C

Melting point: -41.6°C

Relative density (water = 1): 1.4

Solubility in water at 20°C: miscible

Vapour pressure, kPa at 20°C: 6.4

Relative vapour density (air = 1): 2.2

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.07

Octanol/water partition coefficient as log Pow: -0.21

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

Serious local effects by all routes of exposure.

Effects of short-term exposure

The substance is corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. Inhalation may cause asthma-like reactions (RADS). Exposure could cause asphyxiation due to swelling in the throat. Inhalation of high concentrations may cause pneumonitis and lung oedema. See Notes.

Inhalation risk

A harmful contamination of the air can be reached very quickly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure

Repeated or prolonged inhalation may cause effects on the teeth. This may result in tooth erosion. The substance may have effects on the upper respiratory tract and lungs. This may result in chronic inflammation of the respiratory tract and reduced lung function. Mists of this strong inorganic acid are carcinogenic to humans. See Notes.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 2 ppm as TWA; 4 ppm as STEL.

EU-OEL: 2.6 mg/m³, 1 ppm as STEL**ENVIRONMENT****NOTES**

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort. Rest and medical observation are therefore essential.

IARC considers mists of strong inorganic acid to be carcinogenic (group 1). However there is no information available on the carcinogenicity of other physical forms of this substance. Therefore no classification for carcinogenicity under GHS has been applied. NEVER pour water into this substance; when dissolving or diluting always add it slowly to the water.

The odour warning when the exposure limit value is exceeded is insufficient.

Rinse contaminated clothing with plenty of water because of fire hazard.

ADDITIONAL INFORMATION**EC Classification**

Symbol: O, C; R: 8-35; S: (1/2)-23-26-36-45; Note: B

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